

Vulnerability Assessment Report

27th May 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from January 2026 to March 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

The database server is critical to the business because it stores customer orders, payment records, and inventory data. Securing this server ensures the confidentiality and integrity of sensitive information, preventing data breaches that could harm customer trust and business operations. If the server were disabled or compromised, the business could lose access to essential data, interrupt sales, and suffer financial and reputational damage.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Obtain sensitive information via exfiltration	1	3	3
Cybercriminal	Launch ransomware attack to encrypt customer data	2	3	6

<i>Insider(employe e)</i>	<i>Misuse administrative privileges to alter or delete record</i>	2	2	4
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Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

This assessment focuses on threats most relevant to a small business with an online retail database. External attackers and insiders pose the highest risk because they can directly impact data confidentiality and availability. The chosen threats reflect realistic scenarios based on common vulnerabilities in small business environments.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database.

To reduce these risks, the business should implement multi-factor authentication for all administrative accounts, enforce least-privilege access controls, and conduct regular backups to mitigate ransomware impact. Network segmentation and continuous monitoring should be added to detect unusual activity early. Encrypting sensitive data and applying timely software updates will further strengthen defenses.